VIBHAKAR KAUSHIK MOHTA

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EDUCATION

Carnegie Mellon University (CMU) **Pittsburgh, PA**
Master of Science in Robotic Systems Development | TA Robot Mobility and Robot Autonomy | GPA: 4.08/4.0 May 2024
Courses: Visual Learning | Robot Learning | Probabilistic Graph Models | Planning for Robotics | F1Tenth Racing | Computer Vision

Indian Institute of Technology Kharagpur (IIT KGP) **Kharagpur, India**
Bachelors in Mechanical Engineering, Masters in Systems Design | TA Design Optimization | GPA: 9.22/10 May 2022
Minor in Computer Science | Institute Order of Merit recipient for outstanding contributions to technology

SKILLS

Programming Languages: C, C++, Python, C#, SQL, MATLAB
Libraries: PyTorch, Tensorflow, OpenCV, NumPy, SymPy, Pandas, SciPy, STL
Frameworks/Tools: Git, Docker, ROS 1/2, MoveIt 2, MuJoCo, Gazebo, TensorRT, ONNX, Protobuf

EXPERIENCE

Nuro Inc. | Research Scientist, Full-Time **Mountain View, CA** | Jan 2025 - Present
- Post trained flow based behavior models using RL to ensure safe and humanlike driving behavior
- Optimized RL infra and training pipelines to scale to millions of simulated rollouts to learn a general driving policy
- Designing ML architectures to generate kinematically feasible trajectories, ensuring passenger comfort and safety

Plus AI Inc. | Prediction ML Engineer, Full-Time **Santa Clara, CA** | June 2024 - Jan 2025
- Developed a transformer-based architecture to predict joint and conditional distributions of possible vehicle behaviors
- Scaled up training to large-scale on-road data to learn a foundational prediction model, optimizing the model for deployment

Aurora Innovation Inc. | Motion Planning ML Intern **Mountain View, CA** | May 2023 - Aug 2023
- Developed a data-driven metric to analyze and flag undesirable turn signal behavior in high-speed autonomous truck driving
- Evaluated the metric on a large array of on-road and simulation datasets, achieving a ***98.6%*** recall and an accuracy of ***94.2%***

Wadhwani Institute for Artificial Intelligence | Machine Learning Research Intern **Mumbai, India** | May 2021 - Jul 2021
- Implemented ensemble-based uncertainty estimation, obtaining a calibration error of 2% on underweight birth detection
- Project got the **best paper award** at the *Computer Vision for Physiological Measurement* (CVPM) workshop at **CVPR 2024** | Paper

PROJECTS

SAILOR: Searching Across Imagined Latents Online for Recovery | Prof. Sanjiban Choudhury **Cornell** | Sept 2024 - Jan 2024
- Proposed a Learning to Search method for learning to recover from mistakes using planning at test time | **NeurIPS'25 Spotlight**
- Learned a world model, a reward, and a critic network, utilizing latent imagination to propose corrections to base policy plans
- Improved over diffusion policy-based baselines by ***2-3x*** across 12 tasks from 3 different suites | Website | Podcast | Lecture@CMU

Zero-shot Sim2Real model based RL for autonomous racing | Prof John Dolan **CMU** | Feb 2024 - Apr 2024
- Augmented DreamerV3 for autonomous racing, utilizing map randomization and LQR costs to learn deployable racing policies
- Deployed learned policies on a real 1/10 scale race car over ROS 2, demonstrating zero-shot sim2real transfer | Report | Video

Autonomous construction on lunar-like terrain | Prof. Red Whittaker, accepted at NASA LSIC **CMU** | Jan 2023 - Dec 2023
- Prototyped an autonomous excavator that uses continuous excavation to maintain low-ground reaction forces | Website | Github
- Implemented a bilevel optimization algorithm to minimize total operation energy, accounting for worksite constraints | Report

Diffusion policies for manipulation in partially visible environments | Prof. Deepak Pathak **CMU** | Oct 2023 - Dec 2023
- Augmented diffusion policies to multimodal sensor inputs from vision, audio, and joints of the Franka Emika Robot | Report
- Compared performance with standard behavior cloning, obtaining a ***1.6x*** boost on cloning a pick and place expert policy

Improving stable diffusion for human image generation | Prof. Deepak Pathak **CMU** | Feb 2023 - Apr 2023
- Improved stable diffusion's human images by guiding a frozen model with an implicitly conditioned ControlNet branch | Report
- Achieved a ***5%*** decrease in FID, and significant visual improvements over stable diffusion outputs on the MS COCO dataset

Autonomous Jenga block stacking robot | Prof. Oliver Kroemer **CMU** | Feb 2023 - Apr 2023
- Devised a robust manipulation pipeline with recovery behaviors for building Jenga towers using the Franka Emika robot | GitHub
- Achieved a pickup accuracy of ***84%***, employing a heuristic-based approach to minimize poor grasps and select viable blocks

ACHIEVEMENTS

Feb 2026: Runners up at the Physical AI Hack among 45+ teams and 900+ people at Founders, Inc | Post
Oct 2025: Won the best self-improving agent award at Weavehacks2 among 67 teams organized by W&B and Google Cloud | Post
Aug 2025: Featured on Robopapers podcast to talk about world models, reward learning and test time planning | Post